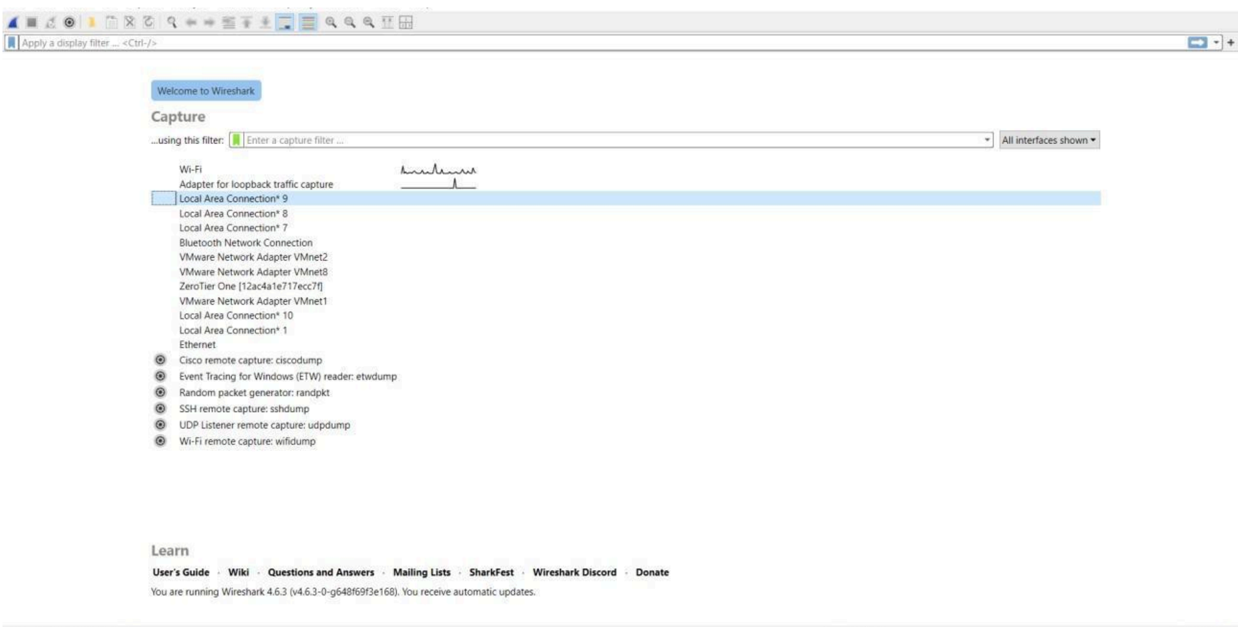
Advanced Computer Networks – PCS223

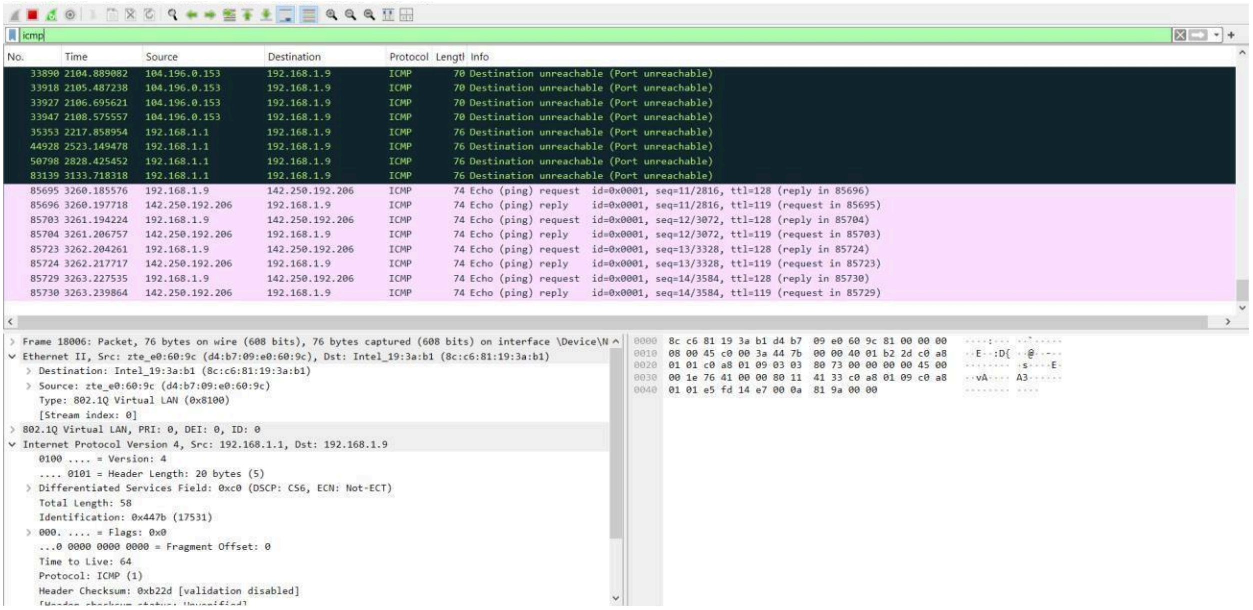
Assignment 2

Name – Taranpreet Singh
Subgroup – CS4
Roll No. – 8025320103

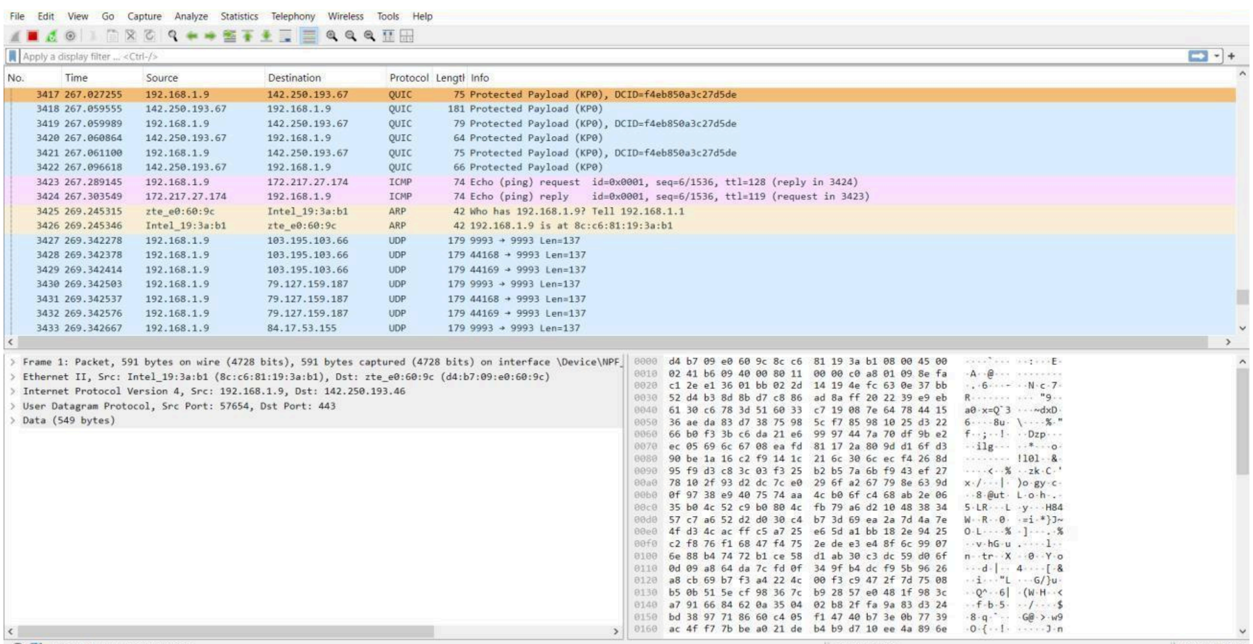


Screenshot showing Wireshark Home screen showing all interfaces and the active interface (Wi-Fi)

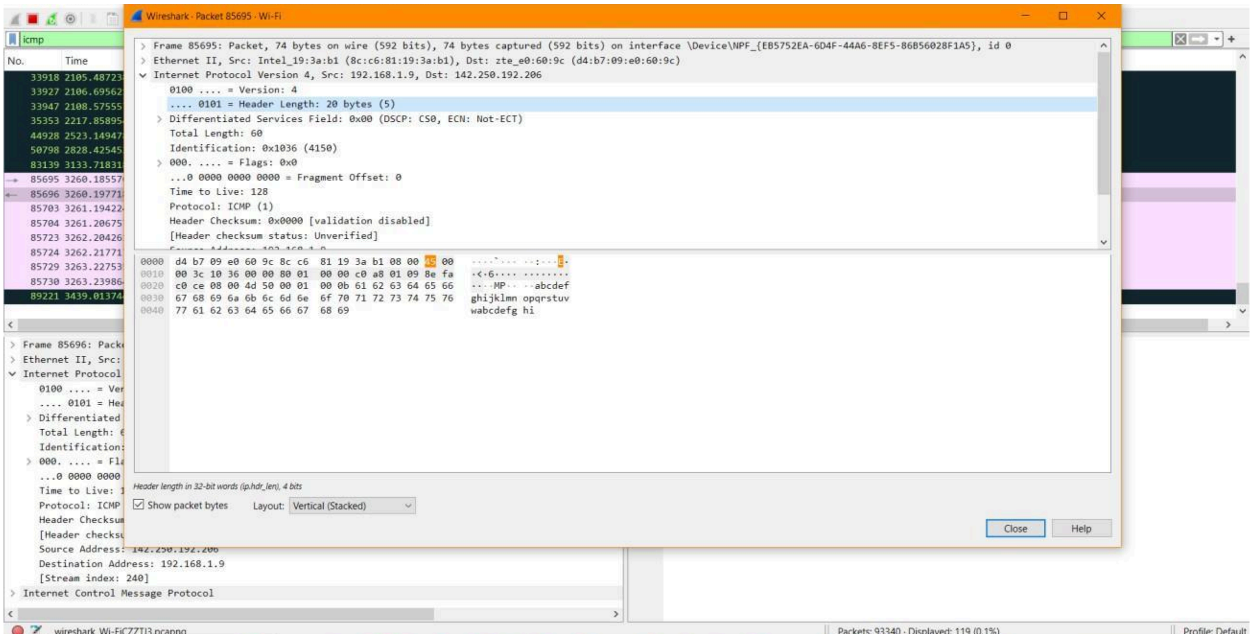
1. To capture and analyze ICMP packets (Ping request and reply)



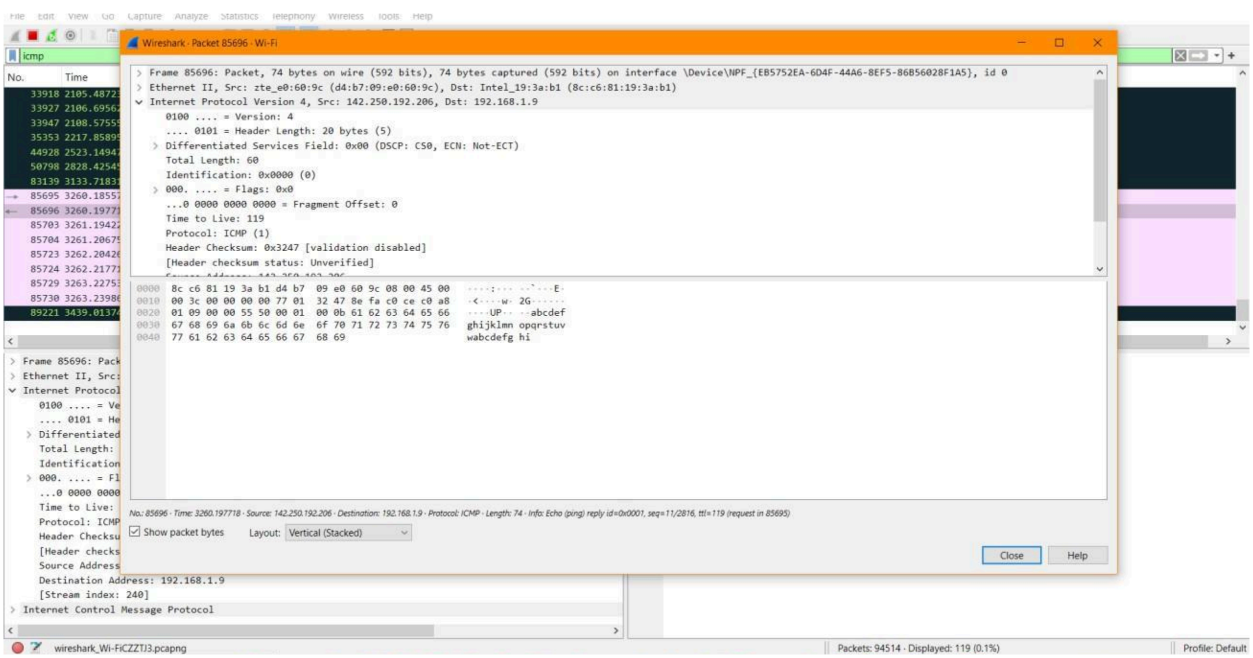
Screenshot showing ICMP Packets Captured by pinging google.com



Screenshot showing ICMP Packets Captured by pinging google.com



Screenshot showing ICMP Packet 85695 details (Ping Request)



Screenshot showing ICMP Packet 85696 details (Ping Reply)

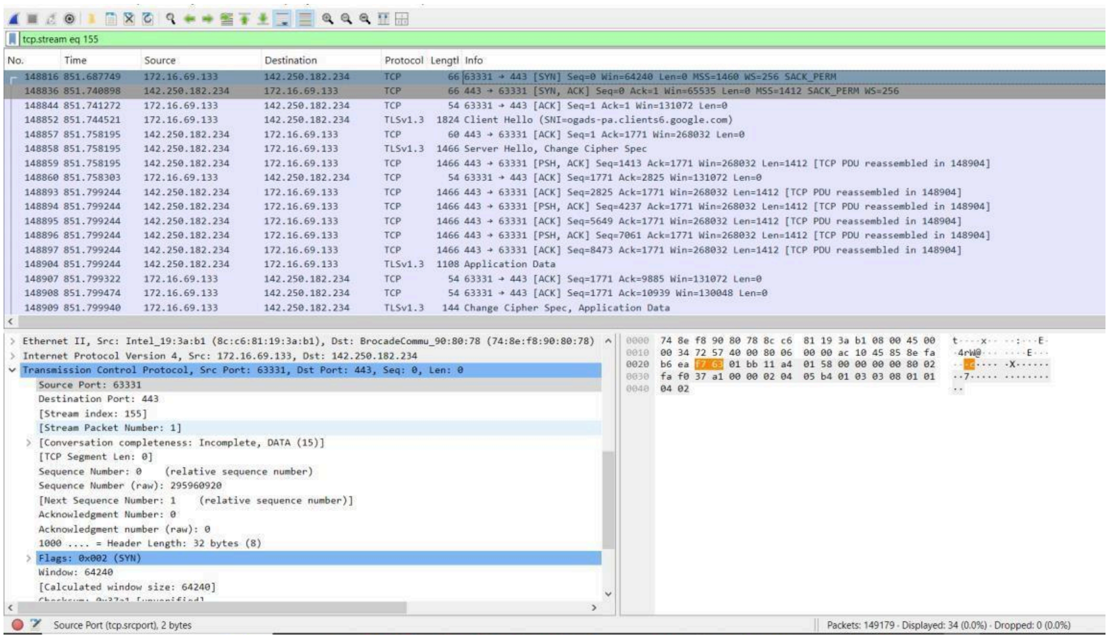
Observation and Analysis of ICMP Packet 85695 -

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and Internet Control Message Protocol
- Source IP Address: 192.168.1.9
- Destination IP Address: 142.250.192.206
- Packet Length: 74 Bytes

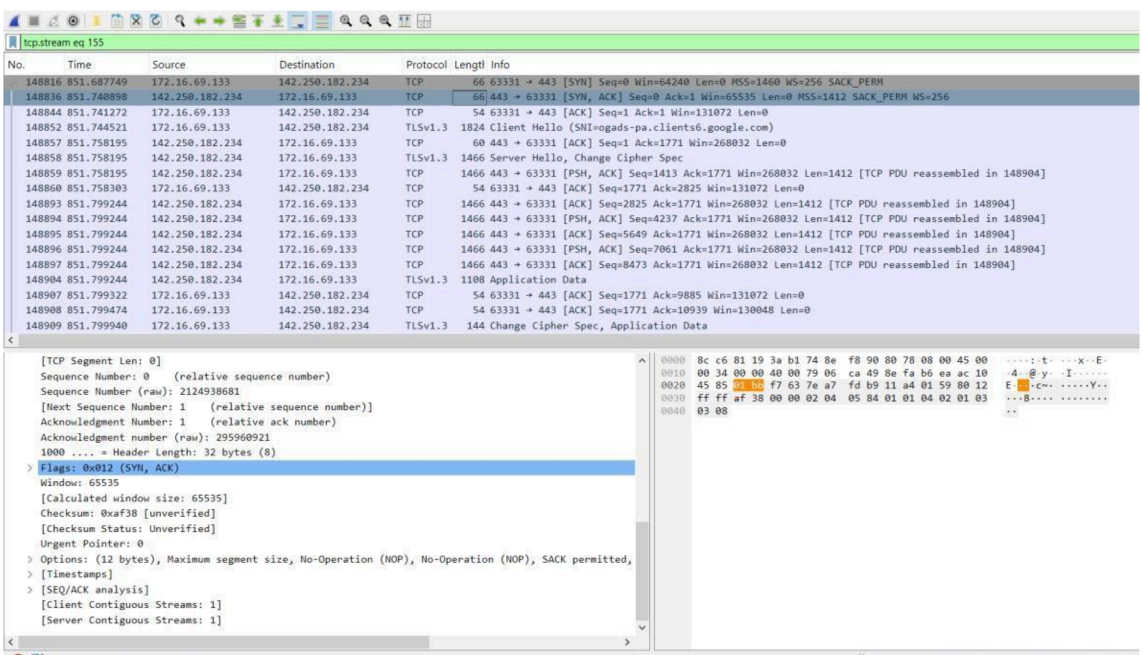
Observation and Analysis of ICMP Packet 85696 -

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and Internet Control Message Protocol
- Source IP Address: 142.250.192.206
- Destination IP Address: 192.168.1.9
- Packet Length: 74 Bytes

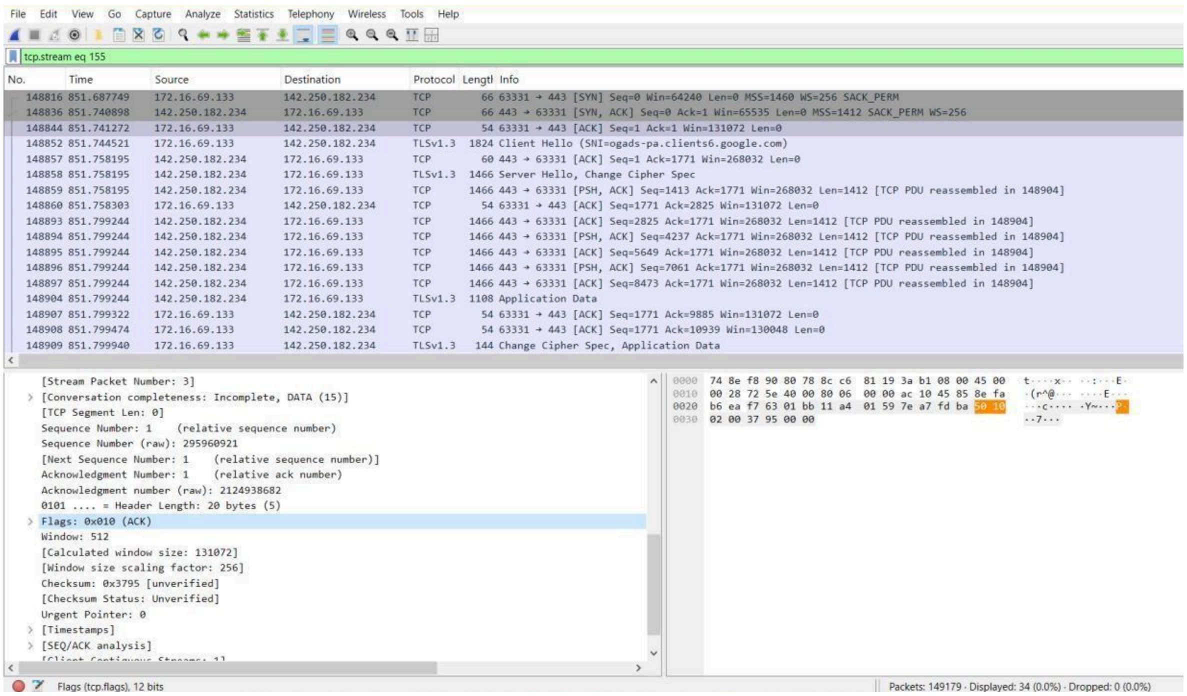
2. To analyze TCP packets and observe the TCP three-way handshake



Screenshot showing TCP Packet (SYN)

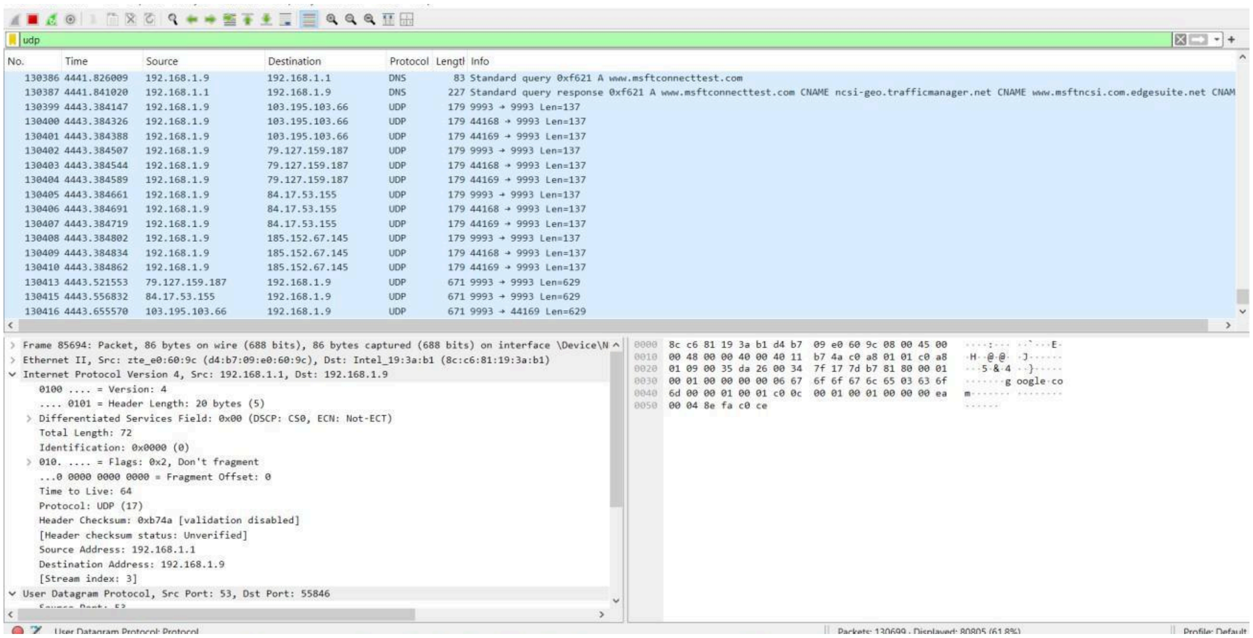


Screenshot showing TCP Packet (SYN, ACK)

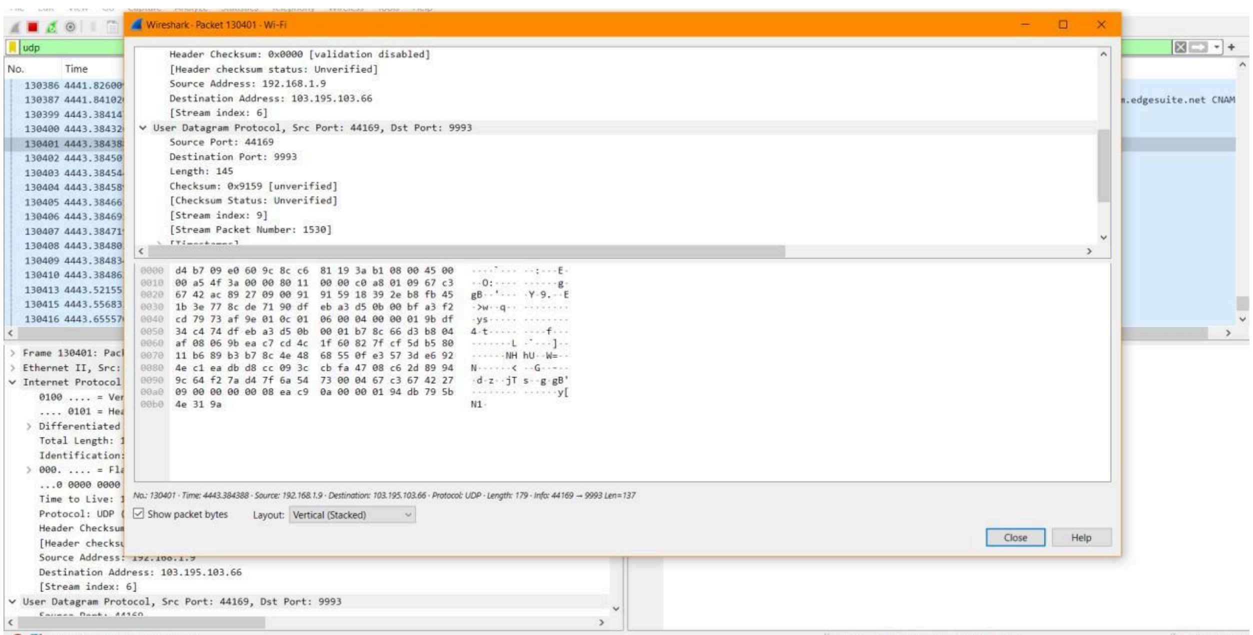


Screenshot showing TCP Packet (ACK)

3. To capture and analyze UDP packets



Screenshot showing UDP Packets Capture

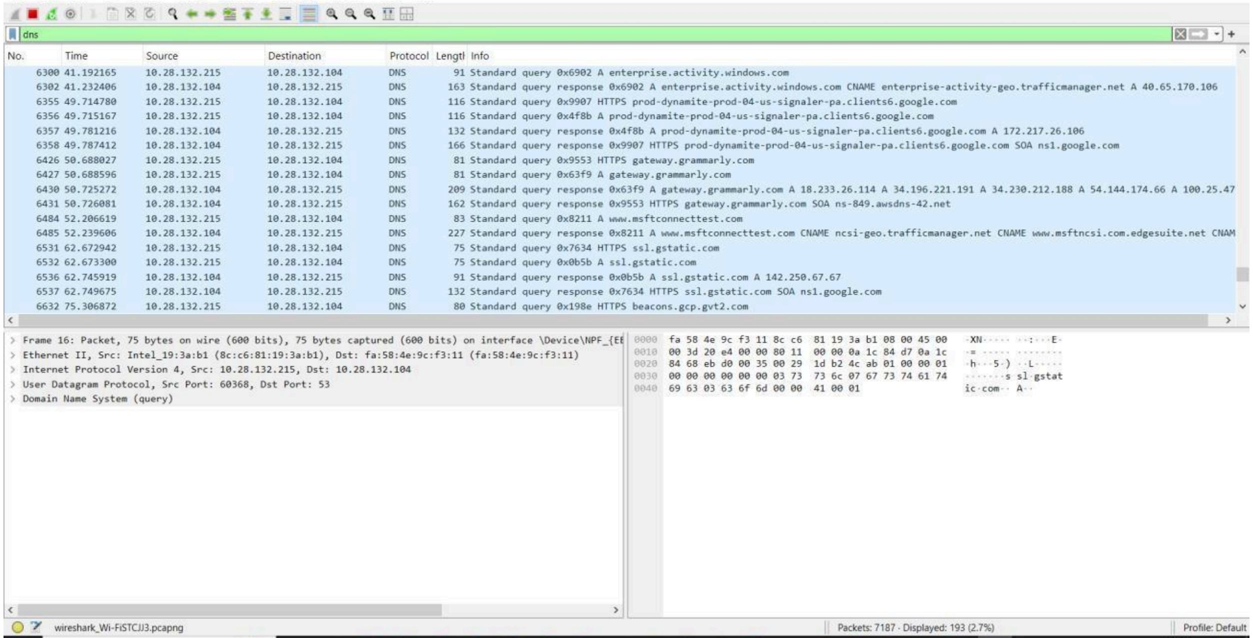


Screenshot showing details of a UDP Packet

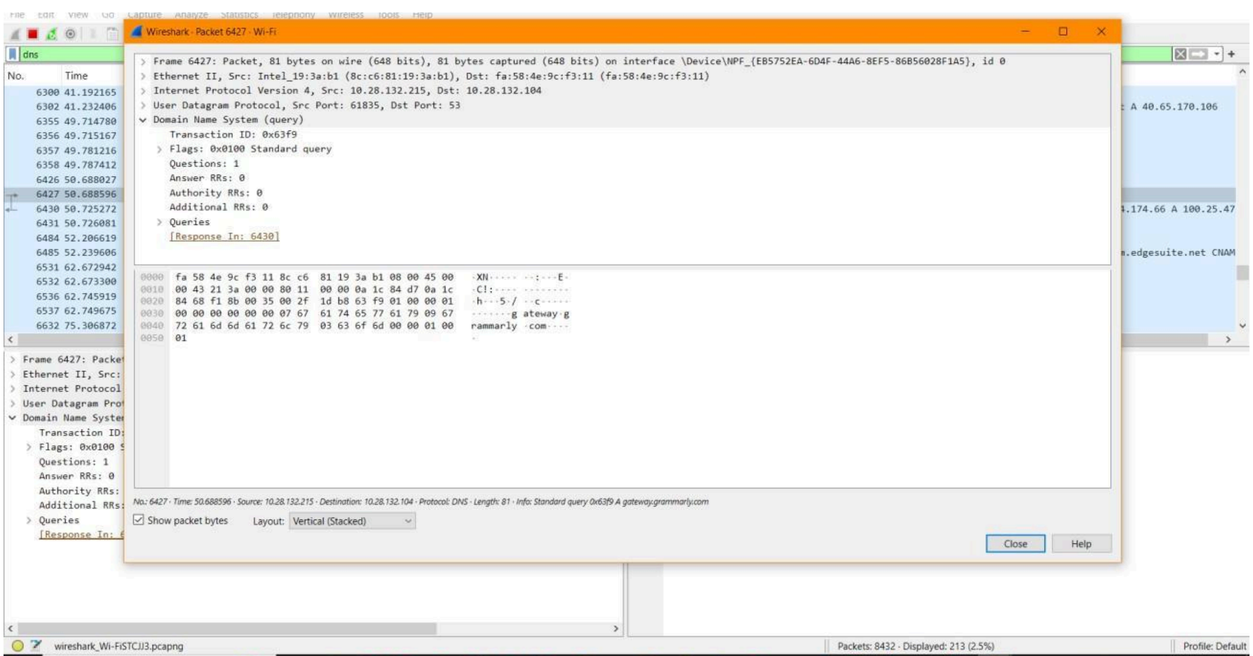
Observation: UDP Packet 130401 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 192.168.1.9
- Destination IP Address: 103.195.103.66
- Packet Length: 179 Bytes

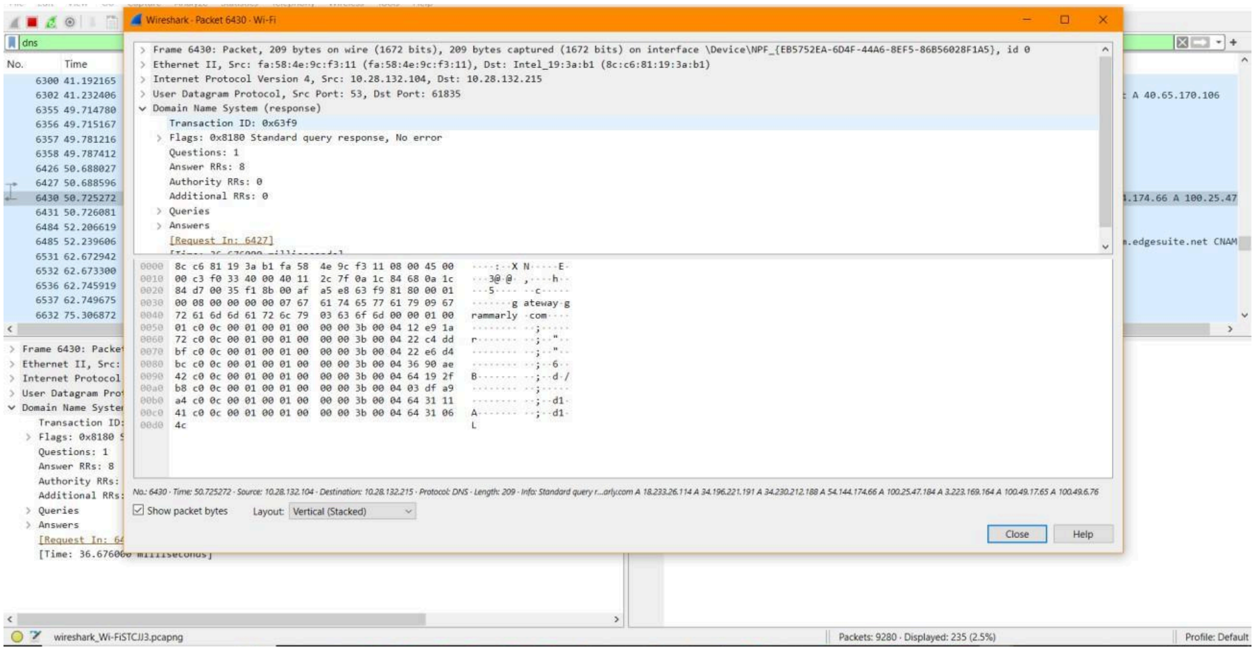
4. To capture and analyze DNS query and response packets



Screenshot showing DNS Packets Capture



Screenshot showing details of a DNS Packet 6427 (Standard query request)



Screenshot showing details of a DNS Packet 6430 (Standard query response)

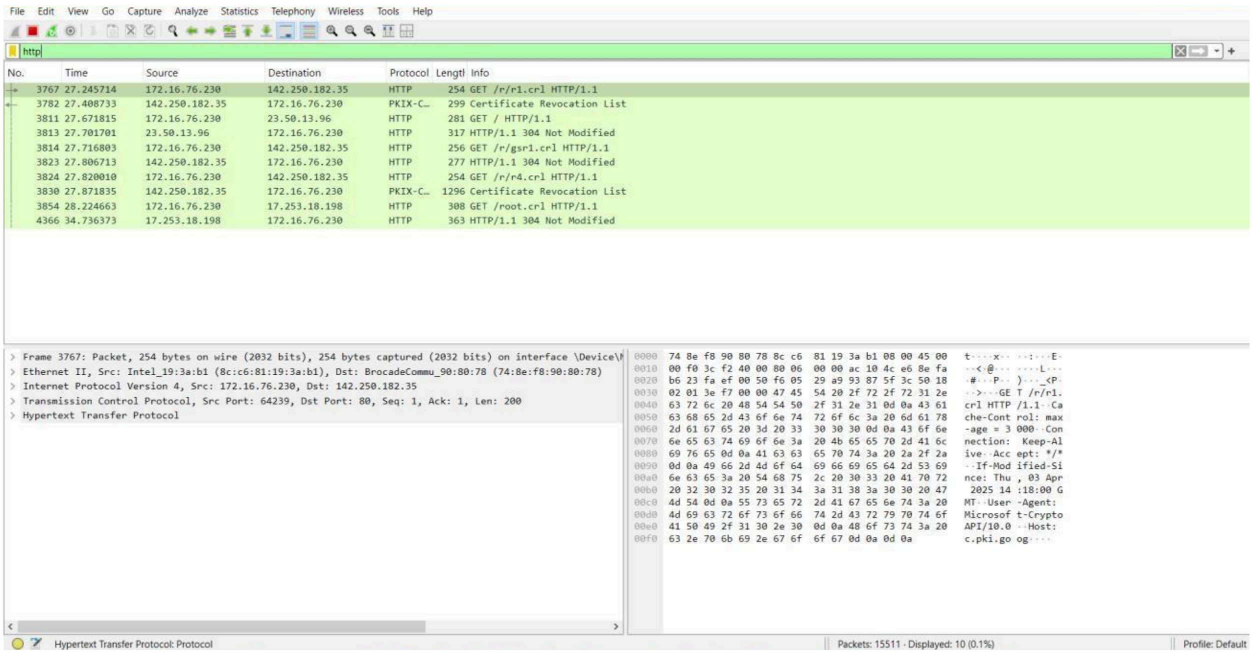
Observation: DNS Packet 6427 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 10.28.132.215
- Destination IP Address: 10.28.132.104
- Packet Length: 81 Bytes
- Queries - gateway.grammarly.com: type A, class IN

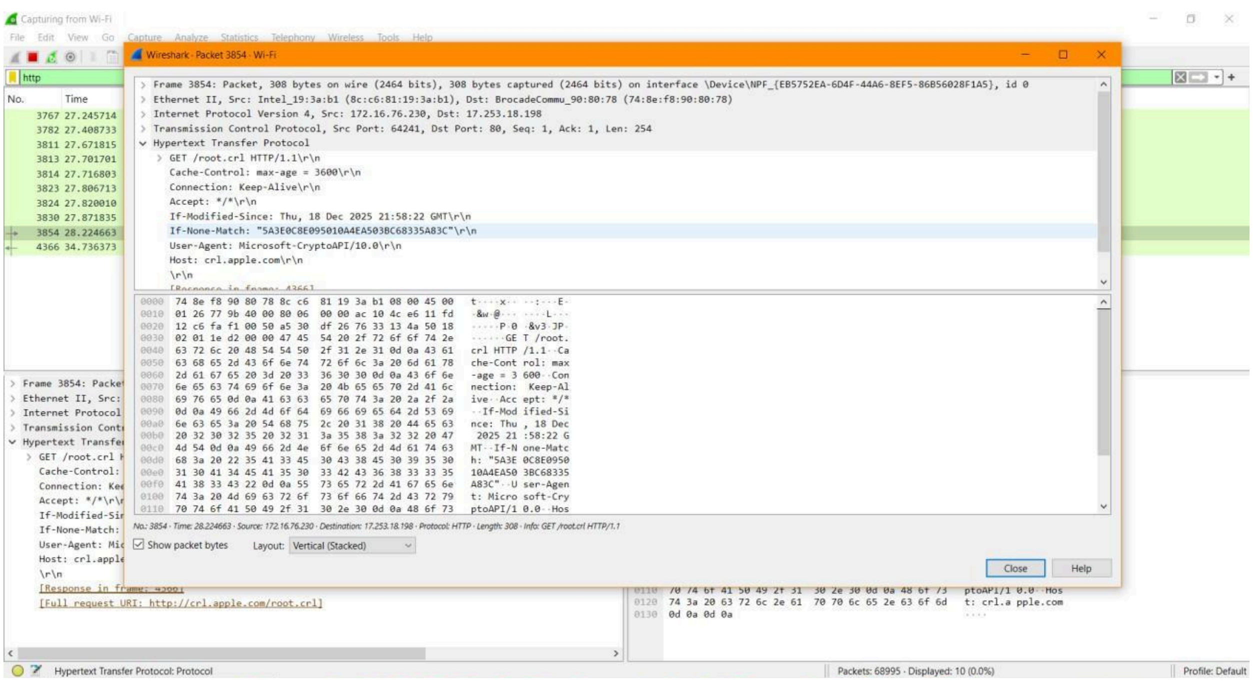
Observation: DNS Packet 6430 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, and User Datagram Protocol
- Source IP Address: 10.28.132.104
- Destination IP Address: 10.28.132.215
- Packet Length: 209 Bytes

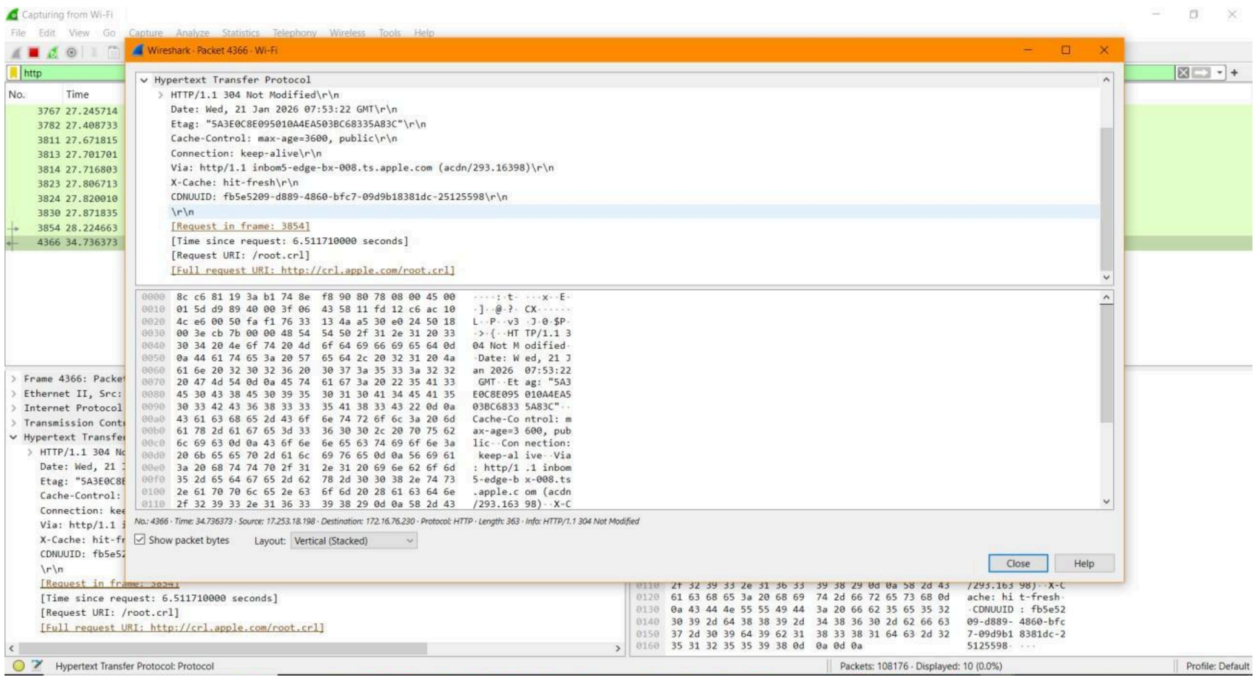
5. To capture and analyze HTTP packets



Screenshot showing HTTP Packets Capture



Screenshot showing details of a HTTP Packet (HTTP Request)



Screenshot showing details of a HTTP Packet (HTTP Response)

Observation: HTTP Request 3854 –

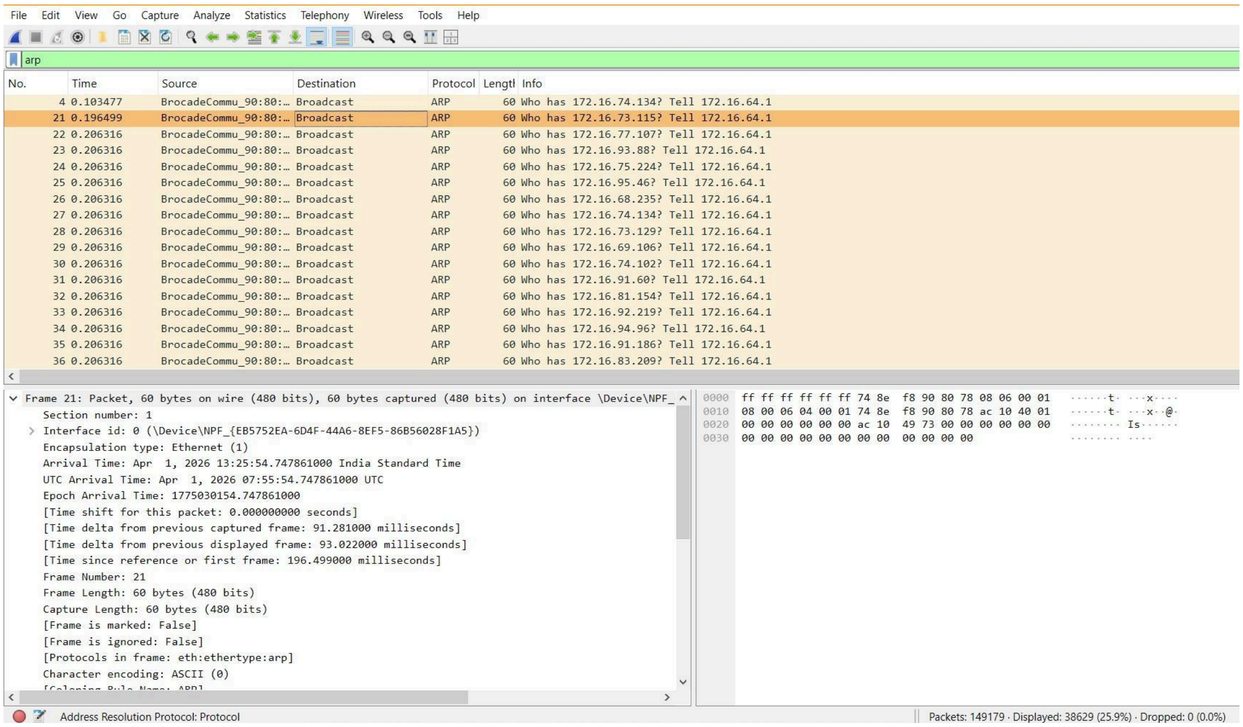
- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, Hypertext Transfer Protocol
- Source IP Address: 172.16.76.230
- Destination IP Address: 17.253.18.198
- Packet Length: 308

Bytes Observation: HTTP Response

4366 –

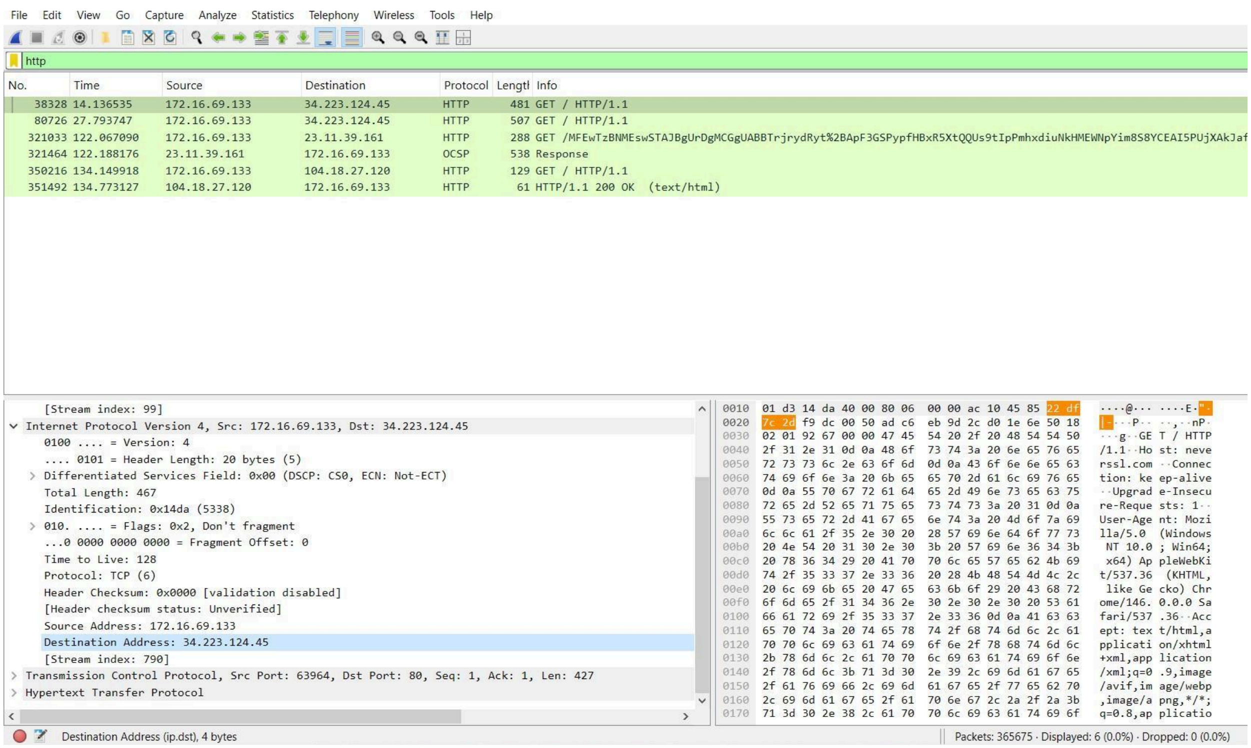
- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, Hypertext Transfer Protocol
- Source IP Address: 17.253.18.198
- Destination IP Address: 172.16.76.230
- Packet Length: 363 Bytes

6. To analyze ARP request and reply packets

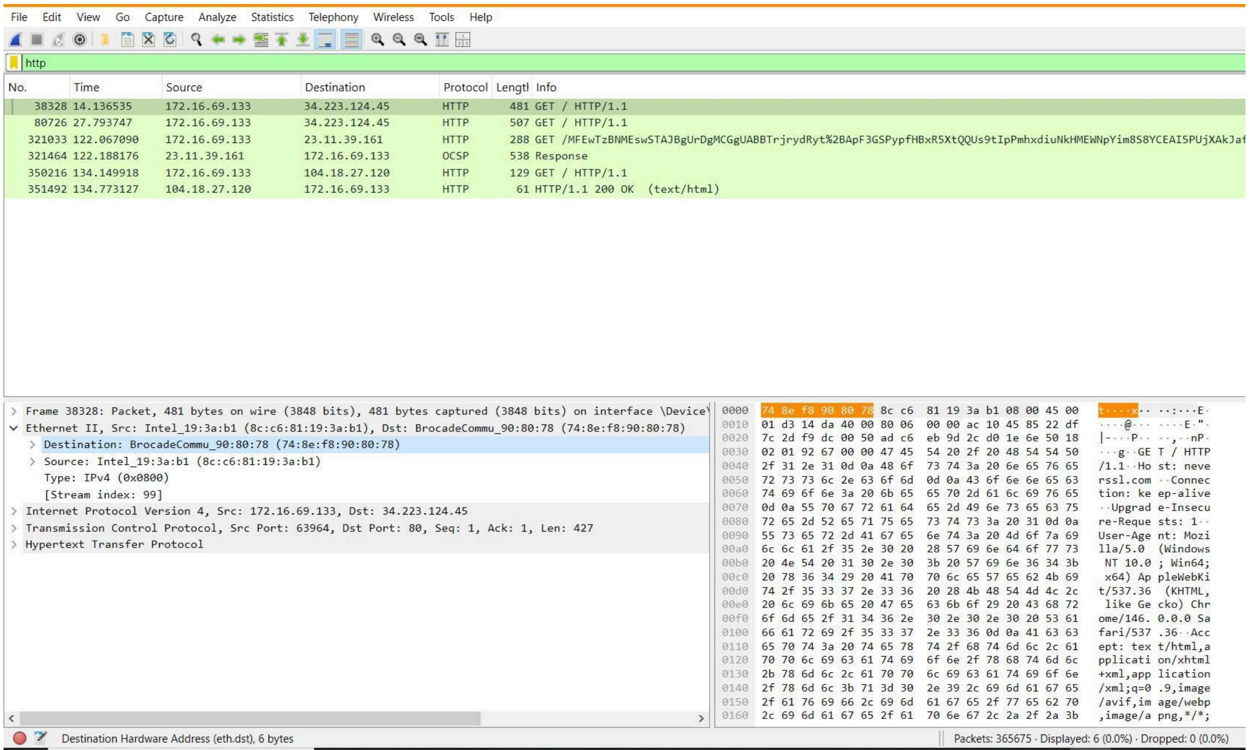


Screenshot showing ARP Packets Capture

7. To identify source and destination MAC and IP addresses in captured packets



Screenshot showing Source and Destination IP Address of a HTTP



Screenshot showing Source and Destination MAC Address of a HTTP Packet

Observation: HTTP Packet 38328 –

- Interface Used: Wi-Fi
- Protocols Observed: Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, Hypertext Transfer Protocol
- Source IP Address: 172.16.69.133
- Destination IP Address: 34.223.124.45
- Source MAC Address: 8c:c6:81:19:3a:b1
- Destination MAC Address: 74:8e:f8:90:80:78
- Packet Length: 481 Bytes